Chapter 1

Monte Carlo Methods

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1 Introduction

In this chapter, the Monte Carlo methods are discussed. Unlike DP, Monte Carlo does not require a perfect MDP model. Monte Carlo learns from experience, or simulated experience, if a model is ready.

Monte Carlo methods are ways of solving reinforcement problem based on averaging sample returns. Unfortunately, the Monte Carlo method can be used only on episodic tasks.

2 Monte Carlo Prediction

When a policy is given, we can estimate a state-value function of the policy by **Monte Carlo prediction**. Monte Carlo prediction averages the sample value of desired states and use it as the estimate of state-value function. The averaging is done when each episode terminates, unlike ordinary DP method which updates in each step. This idea of sampling and averaging the returns underlies all Monte Carlo methods.

Precisely, Monte Carlo prediction are devided into two methods; *first-visit MC method* and *every-visit MC method*. The former one estimates $v_\pi(s)$ as the average of the returns first visits to s. The latter one averages the returns following all visits to s. These two MC methods are very similar but have slightly different theoretical properties.

Algorithm 1: First-Visit MC Prediction, for estimating v_π

Input : π .... a given policy
Output: $V \simeq v_\pi$ .... an estimate of v_π

`// initialize`
$V(s) \in \mathbb{R} \leftarrow$ arbitrary values, for all $s \in \mathcal{S}$
$R(s) \leftarrow$ empty list, for all $s \in \mathcal{S}$
repeat
  generate an episode following π: $S_0, A_0, R_1, S_1, A_1, \ldots, S_{T-1}, A_{T-1}, R_T$
  $G \leftarrow 0$
  foreach step of episode, $t = T-1, T-2, \ldots, 1, 0$ **do**
    $G \leftarrow \gamma G + R_{t+1}$
    `// in every-visit MC method, this if statement is ignored`
    if S_t did not appear in $S_0, S_1, S_2, \ldots, S_{t-1}$ **then**
      append G to $R(s)$
      $V(S_t) \leftarrow average(R(s))$
until convergence

Due to the law of large numbers, the first-visit MC method and every-visit MC method converges to $v_\pi(s)$ as the number of visits to s goes to infinity.

Note that the MC method does not bootstrap; DP method uses values of other states when estimating a value of a given state. MC does not. Due to this property of MC method, MC method is useful when one requires only values of a subset of states.